



Assessing the quality of reflection in student journals: a review of the research

Janet E. Dymont & Timothy S. O'Connell

To cite this article: Janet E. Dymont & Timothy S. O'Connell (2011) Assessing the quality of reflection in student journals: a review of the research, *Teaching in Higher Education*, 16:1, 81-97, DOI: [10.1080/13562517.2010.507308](https://doi.org/10.1080/13562517.2010.507308)

To link to this article: <https://doi.org/10.1080/13562517.2010.507308>



Published online: 17 Jan 2011.



Submit your article to this journal [↗](#)



Article views: 4972



View related articles [↗](#)



Citing articles: 31 View citing articles [↗](#)

Assessing the quality of reflection in student journals: a review of the research

Janet E. Dymen^{a*} and Timothy S. O'Connell^b

^a*Faculty of Education, University of Tasmania, Hobart, TAS, Australia;* ^b*Brock University, St Catharines, ON, Canada*

(Received 23 November 2009; final version received 22 June 2010)

In this paper, we review 11 research articles that examine the level of reflection found in student journals in higher education across a range of disciplines. Our review reveals little to no consistency in the research community around the mechanisms and process of assessing levels of reflection in student journals. Our analysis also reveals that the quality of reflection found in student journals varies considerably across studies. In terms of trends, the five of 11 studies found that students or journals were not overly reflective and contained mostly descriptive accounts of events. Four of the 11 studies found that the majority of students or journals were somewhat reflective. Only two of the 11 studies found a high percentage of students or journals were highly reflective. We conclude with a series of factors to be considered when assessing the quality of reflection in student journals.

Keywords: reflective journals; assessment; levels of reflection; evaluation

Introduction

Dewey (1933) is credited with suggesting that reflection is an important component of learning and theorised that reflection is necessary to incorporate experiences into an existing framework of knowledge, while taking into consideration a learner's life experience as well as present observations. Dewey defined reflection as, '...active, persistent and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it and the further conclusion to which it tends' (9). Since Dewey first introduced his idea and definition of reflection, there has been ongoing debate as to its various components and how reflection actually happens (Iser 1999). Several authors have attempted to clarify reflection through offering a conceptual framework, or model, of reflection (e.g. Boud, Keogh, and Walker 1985; Merizow 1981; Schön 1983).

The process of reflection in higher education can be facilitated in a number of ways. Formal academic papers, questionnaires, workshops, exams, BlackBoard discussion groups and tutorials have all been used to encourage university students to take ownership of their knowledge and to make connections between the theory and practice of their studies. Journal writing as a means of reflection is another popular technique used by educators in many higher education contexts (Jarvis 2001). Journal writing can take many forms and can occur in a classic leather bound

*Corresponding author. Email: janet.dymen@utas.edu.au

book, in a highly structured work booklet, or even an on-line Blog. Journals can also vary in terms of their purpose, with some journals containing merely a listing of 'evidence' with very little analysis/reflection while others contain highly personal and reflective account of events (Moon 2006).

Irrespective of their form and purpose, written journals can serve as a vehicle for reflection before, during and after a learning experience. They can provide a venue for students to develop personally and professionally. Through journals, students can enhance their learning by making connections between theory and practice, by asking questions, and by engaging with higher order ideas (Connor-Greene 2000; Kerka 1996). Unsurprisingly, reflective journals have been embraced across a range of discipline areas in higher education, including pre-service teacher training programmes (Minott 2008; Spalding and Wilson 2002), early childhood education programmes (Cornish and Cantor 2008), nursing programmes (Richardson and Maltby 1995; Thorpe 2004; Wong et al. 1995), as well as physical therapy programmes (Plack et al. 2005; Williams and Sundelin 2000).

Reflective journals are popular in higher education because they allow students to examine their beliefs, values, experiences and assumptions about the subject matter at hand (Minott 2008). Through the reflective practice of journal writing, university students can begin to contextualise their experiences in terms of the relationship of these experiences to their current academic and future professional life. Additionally, and more importantly, they can move beyond the basic ability to recall facts and knowledge, and move towards 'connected learning', or the skill to actively and critically analyse knowledge in different environments (Connor-Greene 2000).

The notion that journals *can* be an effective medium for facilitating deep reflection for higher education students is undeniable. However, researchers have also identified a number of concerns about their use (Dymont and O'Connell 2010). It seems that too often students are simply handed a journal and asked to write about their practical experiences with little or no structure provided (Mills 2008). Educators who use journals often do not provide training in effective journaling (O'Connell and Dymont 2003). Furthermore, many educators do not actively confront the potential pitfalls of using journal writing as a pedagogical method. Some of these issues revolve around complex ethical dilemmas that need to be addressed, include instructors reading personal revelations written by students, assigning a grade to a journal and not allowing adequate time for feedback (English 2001). A number of troubling studies have reported that students feel that they are 'journalled to death' (Anderson 1992, 302), that they are 'journaling for a grade' (Chandler 1997, 46) and that journals are 'a pointless ritual wrapped in meaningless words' (Shor 1992, 83). It appears that student journals in higher education can fail to live up to their potential in helping to facilitate deep reflection.

The ultimate hope, of course, is that journals in higher education will help to facilitate deep and critical reflection. Yet, there is inconclusive evidence as to whether this hope is actually being realised. The research that systematically evaluates the quality of reflection found in student journals paints an unclear picture. While a number of studies reveal that the majority of student journals contain highly reflective entries (e.g. Sumsion and Fleet 1996; Wessel and Larin 2006), another body of research suggests that the majority of student journals contain merely descriptive accounts of events with limited reflection (e.g. O'Connell and Dymont 2004; Richardson and Maltby 1995).

In this paper, we bring together research studies that have sought to interrogate the quality of reflection found in student journals in higher education. More specifically, we examine 11 research articles that have assessed the quality of student journals and answer the following two questions:

- (1) What processes were used to assess the quality of reflection in student journals?
- (2) What was the quality of reflection found in student journals?

Methods

With a view to identifying studies that should be included in this review, we searched a number of electronic data bases. Our search strategy used keywords such as 'journal(s)', 'journaling', 'journal writing', 'reflective journals', 'reflective writing', 'diary', 'assessment', 'reflection', 'level of reflection', 'evaluation' and 'assessment' to focus the results. Approximately, 80 articles were reviewed for primary sources of research.

Using the following inclusion and exclusion criteria, we limited our sample to 11 research articles from a range of disciplines in higher education. In regards to our rationale for dates of inclusion, there were very few studies that met the *other* inclusion criteria pre-1995 so we simply decided to have this as our boundary for containing our sample. Furthermore, the majority of pre-1995 articles were anecdotal in their provision of evidence regarding assessment of the quality and rigour of student journal entries.

Inclusion criteria:

- articles from 1995 to 2009;
- reflective journals were being used by students as some part of a programme of study;
- researchers used a theoretical model or framework to assess the levels of reflection found in student journals;
- students were undergraduate or graduate; and
- minimum sample of 10 students.

Exclusion criteria:

- articles prior to 1995;
- scholarly, theoretical and conceptual articles;
- studies assessing levels of reflection communicated through a source other than journals (e.g. worksheets, interviews, questionnaires); and
- samples of less than 10 students.

Results

What processes were used to assess the quality of reflection in student journals?

Our review of the literature suggests that there is little agreement among the research community as to how to best assess the quality of reflection in student journals. The

11 research articles used a variety of approaches to analysing the level of reflection found in student journals. In the following section of this paper, we describe the variation we found in our review (see Table 1).

Sample

The samples that were assessed across the 11 research studies varied considerably. The undergraduate students involved in the research studies were enrolled in a range of disciplines, including outdoor recreation ($n = 1$), pre-service teacher education ($n = 2$), physiotherapy ($n = 3$), nursing ($n = 4$) and studio design ($n = 1$). The number of students involved across the studies varied from 15 (Wessel and Larin 2006) to 79 (Sumsion and Fleet 1996) students. Although the gender distribution was not stated in all studies, those that did report revealed that the majority of studies included more women than men (e.g. one man: 29 women in Gulwadi 2009; and seven men: 52 women in O'Connell and Dymont 2004). One sample was even comprised of all women (Wessel and Larin 2006).

The demographics of the various samples stand to influence the quality of reflection found in student journals. Interestingly, the large majority of the programmes profiled could easily be deemed as 'caring' professions (e.g. nursing, teaching and physical therapy). The one exception is the studio design programme (Gulwadi 2009). It is plausible that smaller class sizes might influence the quality of reflection – with smaller, more intimate classes supporting a culture of reflection. Although we can not make a claim about the relationship between the number of students involved in a study and the class size, it would be interesting to know if there is a relationship. In terms of gender distribution, there is evidence to suggest that men and women have different perceptions of and attitudes towards journal writing (Burt 1994; O'Connell and Dymont 2003). Women are often reported to be more open to reflective journal writing – so gender distribution might well be an important factor that influences the quality of reflection found in these 11 studies.

Unit of analysis

There was little to no consistency with respect to the unit of analysis. Some researchers assessed student journals once (e.g. O'Connell and Dymont 2004), whereas others collected the journals on multiple occasions and noted trends over time (e.g. Jensen and Joy 2005; Wessel and Larin 2006). The number of items (e.g. journals, entries) assessed varied considerably across the 11 research articles. The largest number of entries was assessed by Richardson and Maltby (1995), who examined 1198 journal entries from 30 student diaries. Minott (2008) looked at far fewer journal entries, examining only five entries from each of 20 students. In terms of actual analysis, some researchers evaluated level of reflection of individual words, sentences and paragraphs (e.g. Plack et al. 2005; Thorpe 2004), whereas others evaluated discrete entries (e.g. Gulwadi 2009; O'Connell and Dymont 2004).

With respect to the reporting levels of critical reflection, some researchers reported the number or percentage of *students'* levels of reflectivity (Thorpe 2004; Wong et al. 1995). Others reported the number or percentage of *entries* (Gulwadi 2009; O'Connell and Dymont 2004) or *journals* (Plack et al. 2005) that achieved a given level of criticality.

Table 1. Selection of research examining quality of journal entries using a range of typologies.

Study and Category	Sample	Description of unit of analysis	Inter-rater reliability	Typology for evaluating journals	Levels/categories (from least reflective to more reflective)
Jensen and Joy (2005) – Low	20 nursing students; five males, 15 females; USA	573 journal entries from 60 journals; collected three times during a semester	Two assessors; $r = 0.63$	Merizow's (1981) model; seven levels of reflection	Percentage of <i>entries</i> at each level (specific data not reported for each level): (1) reflectivity; (2) affective reflectivity; (3) discriminant reflectivity; (4) judgmental reflectivity; (5) conceptual reflectivity; (6) psychic reflectivity; and (7) theoretical reflectivity. 'Lower levels of reflection predominated in journal entries at all three time points – 82% of entries were at levels 1-4' (141).
Minott (2008) – Low	20 pre-service teachers; gender not stated; Cayman Islands	Journals kept during a school term; five entries/ student submitted for assessment	Number of assessors not provided; inter-rater reliability not mentioned	Valli's Typology (Valli 1997)	Number of <i>entries</i> at each level: (1) technical reflection (not assessed); (2) reflection in action/on action (68); (3) deliberative reflection (10); (4) Personalistic reflection (23); and (5) critical reflection (3).
O'Connell & Dymont (2004) – Low	59 outdoor recreation students; seven males, 52 females; Canada, USA	Journal kept during a semester; 880 journal entries from all the students	Two assessors; $r = 0.86$	Bloom's Taxonomy of Cognitive Thinking (Bloom 1956)	Percentages of <i>entries</i> at each level: (1) knowledge (23.33%); (2) Comprehension (37.2%); (3) application (13%); (4) analysis (6.4%); (5) synthesis (7.7%); and (6) evaluation (8.2%).

Table 1 (Continued)

Study and Category	Sample	Description of unit of analysis	Inter-rater reliability	Typology for evaluating journals	Levels/categories (from least reflective to more reflective)
Richardson and Maltby (1995) – Low	30 nursing students; gender not stated; Australia	Random selection of 30 journals from students who had consented to participated in study; 1198 journal entries evaluated	Two assessors; inter-rater reliability was 'Established by exchanging the analysed diaries between the two researchers' (236)	Powell's (1989) tool for the measurement of reflectivity based on the work of Schon (1983) and Merizow (1981)	Percentage of <i>entries</i> at each level: <i>Consciousness</i> (1) reflectivity (36%); (2) affective reflectivity (28%); (3) discriminant reflectivity (30%); and (4) judgmental reflectivity (1%). <i>Critical consciousness:</i> (5) conceptual reflectivity (2%); and (6) theoretical reflectivity (3%).
Wessel and Larin (2006) – Low	15 physiotherapy students; all female; United Arab Emirates	Journals kept during first (4 week) and third (6 week) clinical placements; students required to write one entry per week, submit weekly	Two assessors; worked together to rate level of reflection for each entry	Modified version of Boud et al.'s model (1985); five levels of reflection	Percentage of <i>entries</i> at each level from first to third clinical placement: (1) describes learning (41–22%); (2) analyses learning (20.5–49%); (3) verifies learning (32–21%); (4) gains a new understanding (6–9%); and (5) indicates future behaviour (none).
Gulwadi (2009) – Moderate	30 design students; 29 females, one male; USA	Journals kept for one semester; 226 on-line journal entries	Two assessors; inter-rater reliability not mentioned	Hatton and Smith's Framework (1995); four levels of reflection	Percentage of <i>entries</i> at each level: (1) descriptive (non-reflective) (2.2%); (2) descriptive reflection (38.9%); (3) dialogic Reflection (52.2%); and (4) critical reflection (6.6%).

Table 1 (Continued)

Study and Category	Sample	Description of unit of analysis	Inter-rater reliability	Typology for evaluating journals	Levels/categories (from least reflective to more reflective)
Plack et al. (2005) – Moderate	27 physical therapy students; 13 females, 14 males; USA	Journals kept during 3 weeks of an eight-week clinical experience; total of 48 journals from the 27 students were assessed	Three assessors; inter-rater reliability established through interclass correlational coefficients; results indicated relatively strong to strong agreement	Mixed model based on the work of Merizow (1990), Schon (1987), Brookfield (1987) and Jarvis et al. (1998); three levels of reflection	Percentage of <i>journals</i> at each level: (1) no evidence of reflection (14.7%); (2) Evidence of reflection (43.7%); and (3) evidence of critical reflection (41.9%).
Thorpe (2004) – Moderate	52 nursing students; three males, 49 females; Canada	Journals kept for one semester; worth 25% of final grade	Number of assessors not provided; inter-rater reliability not mentioned; used software to conduct analysis	Scanlon and Chernomas's (1997) model based on Kember et al. (1999) and Wong et al. (1997); three levels of reflection	Number of <i>students</i> at each level: (1) non reflectors (20); (2) reflectors (26); and (3) critical reflectors (6).
Wong et al. (1995) – Moderate	45 nursing students; gender not stated; Hong Kong	One reflective journal/student	Five assessors; $r = 0.88$	Merizow's (1990) model; three levels of reflection	Percentage of <i>students</i> at each level: (1) non-reflective (13.3%); (2) reflective (75.6%); and (3) critically reflective (11.1%).

Table 1 (*Continued*)

Study and Category	Sample	Description of unit of analysis	Inter-rater reliability	Typology for evaluating journals	Levels/categories (from least reflective to more reflective)
Sumsion and Fleet (1996) – High	73 pre-service teachers; gender not stated; Australia	Writing samples were collected four times over a year	Three assessors; $r = 0.50$	Boud et al.'s (1985) model; three levels of reflection	Percentage of <i>students</i> at each level: (1) less-than-moderately reflective/Not reflective (5.4%); (2) moderately reflective (16.4%); (3) highly reflective/more-than-moderately reflective (78%).
Williams et al. (2002) – High	56 physical therapy students; 46 females, 10 males; Canada	Journals kept for one semester; Worth 10% of final grade	Four assessors; each journal read by two of four assessors; 'When coding or grading discrepancy occurred, the instructors discussed the discrepancy and came to a consensus'	Modified version of Boud et al.'s (1985) model; five levels of reflection	Percentage of <i>students</i> at each level: (1) describes learning (0%); (2) analyses learning (2%); (3) verifies learning (23%); (4) gains a new understanding (36%); and (5) indicates future behaviour (39%).

Given all these differences in approaches to analysis, caution is warranted when comparing these findings. It is worthwhile to consider if some degree of consistency in assessment is warranted.

Method of analysis

Whilst not all articles contained a detailed description of the approach to analysis, the following observations can be made. There were varying numbers of assessors involved in the studies – ranging from one (Thorpe 2004) to five (Wong et al. 1995). The majority of studies used two assessors and their experience level varied considerably. There were graduate students, research assistants, faculty members of varying levels and external moderators involved in assessing the level of reflection in student journals. All researchers reported that manual assessment was completed with the exception of Thorpe (2004), who used a qualitative software programme.

Given that the majority of the studies used more than one assessor, the issue of inter-rater reliability became relevant. The amount of information provided varied considerably – some gave very detailed information (e.g. Plack et al. 2005), whereas others gave little to no information on how differences were resolved amongst assessors (Gulwadi 2009). When information was provided, there were varying degrees of reliability, from Sumsion and Fleet (1996) who reported troubling low levels of agreement ($r = 0.50$) to Wong et al. (1995) who indicated highly acceptable levels ($r = 0.88$).

While some researchers used a quantitative system for ascertaining levels of inter-rater agreement, others implemented a qualitative procedure during the data coding process. These researchers independently coded the data, discussed their difference of opinion and reached a joint conclusion on how the entry/journal should be coded (Williams et al. 2002). Others used a slightly different approach in which two researchers coded each journal (Wessel and Larin 2006). Interestingly, this issue of inter-rater reliability indicates one of the key challenges with reflective journal writing assignments in general – what the journal writer means to convey through his or her entry may be interpreted differently than intended, particularly if more than one reader is involved. The diverging inter-rater reliability levels indicated across these studies further highlight this issue.

Models

There was no consistency across articles in terms of the models or theoretical framework used to assess levels of reflection (see Table 1). The models used included the following, or variations on the following:

- Bloom's Taxonomy of Higher Order Thinking (Bloom 1956).
- Valli's Typology (Valli 1997).
- Hatton and Smith's Framework (Hatton and Smith 1995).
- Merizow's Typology (Merizow 1981; Merizow and Associates 1990).
- Boud's Framework (Boud, Keogh, and Walker 1985).
- Powell's Model (Powell 1989).
- Kember et al.'s Model of Reflection (Kember et al. 1999).
- Schön's Reflective Practitioner Framework (Schön 1983, 1987).

The models varied considerably in terms of their complexity – some studies used a model that had only three levels of reflection (e.g. non-reflective, reflective and critically reflective used by Plack et al. 2005), whereas other studies used models that had seven levels (Jensen and Joy 2005). Each model has a range of diverse standards and criteria designed to help determine what constitutes various levels of reflection. Some instruments are clear and straightforward, yet it could be argued that they are overly simplistic. Others are more involved and complicated and whilst they capture more of the richness of the possibilities around reflection, it is plausible that they could be daunting to use given their complexity.

What was the quality of reflection found in student journals?

The results of the 11 studies were categorised in terms of the respective authors' reports on the levels of reflection found in their research. For example, Jensen and Joy (2005) indicated that 82% of the entries were written at the bottom four levels of a seven-level scale. In contrast, Williams et al. (2002) found that 75% of the entries made on a five-level scale were at the two highest levels. Using these reports, we assigned each of the studies to one of three generic categories of reflection: low, medium and high.

It is important to recognise potential issues with categorising the studies in this way. The varying methods used to assign level of reflection, reported issues with inter-rater reliability and unit of analysis used require that these results be considered with a note of caution (see Table 1 for a synopsis of this information).

Low levels of reflection

Using the process described above, five of the 11 studies were categorised as low. These authors reported that the majority of students or journal entries were writing at the 'lower' levels of reflection. By way of example, O'Connell and Dymont (2004) evaluated 880 journal entries from outdoor recreation students and found troubling results: 74% of the entries were written at the lowest three levels of Bloom's Taxonomy (knowledge = 23.3%, comprehension = 37.2% and application = 13%). The remaining entries were distributed evenly over the higher three levels of the taxonomy.

Wessel and Larin's (2006) study was categorised as low as they reported that, on average, students wrote at the lowest three levels of reflection (on a five-level scale), 92.5% of the time. Interestingly, they reported that no entries were made at the highest level, 'indicates future behaviour'. The studies by Minott (2008), Richardson and Maltby (1995) as well as Jensen and Joy (2005) were also included in this category.

Moderate levels of reflection

In terms of the moderate category, it is more straightforward to compare these studies as a similar metric was used in each. Overall, four of the 11 studies included in this research found that students were using their journals to write some degree of reflection, but generally lacking in critical reflection. For example, Wong and colleagues (1995) used Merizow and Associates' (1990) model of reflection to assess

journals and found that 75.6% of their nursing students were being 'reflective' and only 11.1% were considered 'critical reflectors' – the highest level. They surmised that perhaps students were reflecting without recording their thoughts in their journals, or that many students were getting something out of their experiences, but were unable to connect it to other learning opportunities.

Thorpe (2004) used Scanlon and Chernomas' (1997) *stages* of reflection as a foundation for analysing the journals of 52 students. Data analysis were informed by both Kember et al.'s (1999) and Wong et al.'s (1997) *levels* of reflection. Thorpe reported that 50% of students ($n = 26$) were classified as 'reflectors', or the middle of three levels. Likewise, Gulwadi (2009) found that a vast majority (91.4%) of students were writing at the middle two levels of a four-level framework. It is important to note that the students in Gulwadi's study were keeping an on-line journal which may have impacted the level of reflection at which they wrote.

Finally, there was a relatively equal distribution between the middle (43.7%) and highest (41.9%) levels of reflection reported by Plack and colleagues (2005). However, when considering that the remaining 14.7% of journals were written at the lowest level of reflection, this study was categorised into the generic medium category.

High levels of reflection

Disappointingly, only two of the 11 studies found more positive results, whereby the majority of students were being highly critical. Of these two studies, Sumsion and Fleet (1996) found positive results, with 78% of the pre-service teacher journal entries being 'highly reflective'. They found that only 5.4% of their participants were writing at the lowest level of reflection (i.e. less than moderately reflective/not reflective). However, it is important to note that this study was plagued by a number of assessment challenges – including a very low inter-rater reliability ($r = 0.050$) which tempers these findings.

Williams et al. (2002) found that 75% of students were writing at the two highest levels of reflection using a modified framework for reflection based on Boud, Keogh, and Walker's (1985) work. Of these students, 36% showed they were gaining new understanding, and importantly, 39% indicated they would change future behaviour based on their learning.

Discussion

What processes were used to assess the quality of reflection in student journals?

The processes that were used across the 11 studies profiled in this paper vary considerably. More specifically, they differed in terms of the sample, unit of analysis, method of analysis and analytical models. As noted previously, these vastly different approaches make it somewhat difficult to compare and contrast the findings across the 11 studies (hence, we offered our comparison above with due caution). It certainly would be an interesting exercise to assess all the journals using a consistent approach. We think it would be particularly important to have one group of assessors use one theoretical model to assess the journals and embrace a consistent

approach to resolving issues of inter-rater difference. This would ultimately shed insight into the overall quality of the student journals profiled in these 11 studies.

Is it problematic that there are so many approaches used to assess the quality of reflection in student journals? Is there a need for a consistent approach across disciplines? Should a single set of criteria and standards be adopted for assessment of journals across disciplines? We do not have an unequivocal answer to these questions, but we do sense there stands to be some benefit to having some degree of consistency. Given that reflective journals are still seen, by some, to be on the 'edge' or the 'margins' when compared to conventional assessment tasks (O'Connell and Dymont 2006), such as exams and essays, perhaps there is good reason for the academic community to embrace a more consistent approach to assessing the quality of student journals. A consistent approach to assessment across disciplines in higher education might provide an unambiguous signal that this form of learning through journals is recognised, honoured and valued – as much as the other more traditional tasks, like essays and examinations (Crème 2005). We wonder if journals that aren't assessed consistently might be dismissed by students and educators as not being important or legitimate. If journals are seen to be a valuable, then perhaps some degree of consistency is warranted.

We point the readers to the work of Moon (2006) who brought together a number of tools for assessing reflection and has developed what we find to be a fairly comprehensive set of factors, criteria and standards to consider when assessing journals. Perhaps this could be a spring board for a common approach to assessment across disciplines.

Furthermore, a consistent approach to assessment might well influence student learning. It has been argued that what is assessed has a 'powerful influence on student learning' (Crème 2005, 289). Reflective journals require students to engage in critical reflection and higher order thinking; they force students to be more open-ended and less prescriptive; and, they permit students to be creative and questioning. When a journal is assessed, it requires students to open up to these different ways of representing knowledge. For many students, knowing the journal will be assessed is reason enough to persevere with a task, in a way they might not if the journal were not assessed (Crème 2005). Indeed, grades can be a great motivator for some students who might be more reluctant to tap into this way of learning. Although it might be a chore for some students, if educators believe strongly enough in the pedagogical importance of journaling, then perhaps a consistent approach to assessment will ultimately achieve the greatest good for the greatest number of students.

What was the quality of reflection found in student journals?

Given that the literature often espouses the benefits of reflective journal writing, the findings noted in this paper are somewhat troubling – only two of the 11 studies found that the majority of students or entries are highly reflective. Why is this so? How has this come to be?

In this section of the paper, we summarise the key contextual factors that can limit and enable deeply reflective journal writing (see Dymont and O'Connell, 2010 for a full discussion of these factors). These factors emerged from the 11 articles included herein, as well as others that did not meet the inclusion criteria, and provide insight into what is required to help students write more reflective journals.

Were the journals optional or mandatory?

Although it was not always stated, it appears that the majority of journals assignments profiled were mandatory. However, it is possible that some may have been optional. The issue of optional or mandatory is an interesting one (Chandler 1997; Crème 2005). On one hand, optional journals might provide students with a sense of freedom to write more deeply and critically – without the fear of judgement being passed during assessment. On the other hand, mandatory journals that are assessed might invite deeper reflection from those students who are motivated by grades and who might otherwise be less interested in writing critically.

How much were the journals worth in terms of overall grading?

Few studies gave details of how much a journal was worth, but we suspect there might well be a direct relationship between overall worth and the level of critical reflection found in journals (i.e. a journal that is worth more would contain deeper reflection). Williams et al. (2002) as well as O'Connell and Dymont (2004) report that their journals were worth 10% of the final course grade. The highest reported value was 25%, from Thorpe (2004). Some journals were not allocated a percentage, but students were simply required to complete and submit them to pass the course (Wessel and Larin 2006). Grades can be a powerful motivator for some students – and perhaps those worth more would encourage some students to write more deeply and reflectively (Chandler 1997; Crème 2005).

How long and how often were the students required to keep the journals?

In general, students were required to keep the journal as part of a course. Many of the nursing and physical therapy educators required students to keep the journal while on a multi-week clinical placement (Plack et al. 2005; Richardson and Maltby 1995; Wessel and Larin 2006). The outdoor recreation lecturers collected student journals written while students were on a six to nine day field experience (O'Connell and Dymont 2004). The majority of the studies assessed the journals only once; however, some studies collected the journals on multiple occasions and looked for changes in reflection over time (Jensen and Joy 2005; Wessel and Larin 2006). Some studies collected multiple writing samples and did not report on changes over time, but simply on overall levels of reflection (e.g. Sumsion and Fleet 1996). These factors (length of time and collection frequency) might influence the quality of reflection found in student journals (Dymont and O'Connell, 2010). It is possible that students might develop reflective skills over time, with the support of regular feedback and training. Conversely, students might get bored writing journals and their effort to be deeply reflective might decrease over time.

How much training and support was given to the students?

The 11 research articles contained limited information as to the level of training and support provided to students to help them write their journal. O'Connell and Dymont (2004) report that they gave their students a 45-minute workshop on journal writing. Similarly, Plack and colleagues (2005) report that their physiotherapy

students were offered a 1.5-hour training session, whereby students received instruction on the importance of reflective practice in physical therapy and the need to make journal entries. As is postulated in the literature, we wonder if students who did receive training might submit higher quality journals (Anderson 1992; Dymont and O'Connell, 2010; O'Connell and Dymont 2003)?

What kind of feedback response was implemented?

Almost no information was revealed about the kinds of formative and summative feedback that was provided for students. However, this certainly stands to influence the level of reflection found in student journals. As has been noted in the literature, timely and appropriate feedback on student journals can support students to write more reflectively in subsequent journals (Elbow 1997; Fenwick 2001; Hodges 1997). Thus, the students whose journals that were submitted on multiple occasions (e.g. Jensen and Joy 2005; Sumsion and Fleet 1996; Wessel and Larin 2006) might be in a better position to be journaling more deeply and reflectively than their counterparts who were not provided with feedback.

How clear were the instructions around student journaling?

The articles contained a limited amount of information as to the nature of the instructions that were provided around the expectations (e.g. purpose and intent of journal; specific requirements in terms of length, word count, number of entries and role/use of computer?). Some studies provided the exact directions that were provided to students (Wessel and Larin 2006, 122–3; Williams et al. 2002, 9), but this was rare. It has been noted in the literature that educators need to provide very clear directions to students to invite deep reflection (Dymont and O'Connell, 2010; Moon 2006), otherwise they risk having students feeling as though the journal is simply annoying busy work (Mills 2008).

What is the nature of the relationship between the student and educator?

The research articles provide almost no information about the relationships between the journal reader and the journal writer. Yet, it has been noted in the academic literature that this can be one of the key factors that influences the level of deep reflection (Paget 2001). If educators have a trusting relationship with their students, then students might be more open to writing their deep and critical thoughts. Conversely, if a student lacks a trusting relationship with their educator, then they most likely will be reluctant to journal deeply and honestly.

Conclusion

The research that examines the use of reflective journals in higher education presents an unclear picture of the quality of reflection found in student journals. Some research reveals that student journals contain mostly deep reflective entries; other research presents a less optimistic perspective, revealing that most student journals contain mostly descriptive accounts of events, with little to no critical reflection present.

In this paper, we shed further insight into these differing perspectives on the quality of reflection found in student journals. Specifically, we conducted a review of the literature to identify research articles that systematically analysed the level of reflection presented in student journals. We brought together 11 research articles that sought to interrogate the quality of reflection in student journals. In conducting this review, we found there were widely differing approaches being used to assess student journals. We also found that the quality of reflection found in student journals was varied: five studies reported the majority of students or journals lacked in criticality, four studies revealed a moderate degree of reflection by students or in journals and two studies presented more encouraging results, with the large majority of students or journals being highly reflective.

We encourage researchers and educators who are interesting in maximising the potential of journals to seriously consider the findings presented in this paper. In regards to the first research question, perhaps a healthy dialogue is warranted as to the benefits of a consistent approach to assessing reflective journals in higher education. In light of the second research question, we urge educators to reflect upon the generally disappointing findings related to the quality of assessment in student journals. We also invite them to consider and incorporate some of the strategies we presented that stand to enable more critical reflection in student journals.

References

- Anderson, J. 1992. Journal writing: The promise and the reality. *Journal of Reading* 36, no. 4: 304–9.
- Bloom, B.S., ed. 1956. *Taxonomy of educational objectives*. New York, NY: Longmans-Green.
- Boud, D., R. Keogh, and D. Walker. 1985. *Reflection: Turning experience into learning*. London: Kogan Page.
- Burt, C. 1994. An analysis of self-initiated coping behaviour: Diary-keeping. *Child Study Journal* 24, no. 3: 171–90.
- Chandler, A. 1997. Is this for a grade? A personal look at journals. *English Journal* 86, no. 1: 45–9.
- Connor-Greene, P.A. 2000. Making connections: Evaluating the effectiveness of journal writing in enhancing student learning. *Teaching of Psychology* 27, no. 1: 44–6.
- Cornish, M.M., and P.A. Cantor. 2008. “Thinking about thinking: It’s not just for philosophers”: Using metacognitive journals to teach and learn about constructivism. *Journal of Early Childhood Teacher Education* 29, no. 4: 326–39.
- Crème, P. 2005. Should student learning journals be assessed? *Assessment & Evaluation in Higher Education* 30, no. 3: 287–96.
- Dewey, J. 1933. *How we think*. Boston, MA: D.C. Heath.
- Dyment, J.E., and T.S. O’Connell. 2010. The quality of reflection in student journals: A review of limiting and enabling factors. *Innovative Higher Education* 35, no. 3: 233–44.
- Elbow, P. 1997. High stakes and low stakes responding in assigning and responding to writing. In *New directions for teaching and learning: Writing to learn: Strategies for assigning and responding to writing in the disciplines*, Vol. 69, ed. M.D. Sorcinelli and P. Elbow, 5–13. San Francisco, CA: Jossey Bass.
- English, L.M. 2001. Ethical concerns relating to journal writing. In *New directions for adult and continuing education: Promoting journal writing in adult education*, Vol. 90, ed. L.M. English and M.A. Gillen, 27–35. San Francisco, CA: Jossey-Bass.
- Fenwick, T.J. 2001. Responding to journals in a learning process. In *New directions for adult and continuing education: Promoting journal writing in adult education*, Vol. 90, ed. L.M. English and M.A. Gillen, 37–47. San Francisco, CA: Jossey-Bass.
- Gulwadi, G.B. 2009. Using reflective journals in a sustainable design studio. *International Journal of Sustainability in Higher Education* 10, no. 2: 96–106.

- Hatton, N., and D. Smith. 1995. Reflection in teacher education: Towards definition and implementation. *Teaching and Teacher Education* 11, no. 1: 33–49.
- Hodges, E. 1997. Negotiating the margins: Some principles for responding to our students' writing, some strategies for helping students read our comments. In *New directions for teaching and learning: Writing to learn: Strategies for assigning and responding to writing in the disciplines*, Vol. 69, ed. M.D. Sorcinelli and P. Elbow, 77–89. San Francisco, CA: Jossey-Bass.
- Ixer, G. 1999. There's no such thing as reflection. *British Journal of Social Work* 29: 513–27.
- Jarvis, P. 2001. Journal writing in higher education. In *New directions for adult and continuing education: Promoting journal writing in adult education*, Vol. 90, ed. L.M. English and M.A. Gillen, 79–86. San Francisco, CA: Jossey-Bass.
- Jensen, S.K., and C. Joy. 2005. Exploring a model to evaluate levels of reflection in baccalaureate nursing students' journals. *Journal of Nursing Education* 44, no. 3: 139–42.
- Kember, D., A. Jones, A. Loke, J. McKay, K. Sinclair, T. Harrison, C. Webb, F. Wong, M. Wong, and E. Yeung. 1999. Determining the level of reflective thinking from students' written journals using a coding scheme based on the work of Mezirow. *International Journal of Lifelong Education* 18, no. 1: 18–30.
- Kerka, S. 1996. *Journal writing and adult learning*. Technical Report No. 174. Columbus, OH: ERIC Clearinghouse on Adult Career and Vocational Education (ERIC Document ED 339413).
- Merizow, J. 1981. A critical theory of adult learning and reflection. *Adult Education* 32, no. 1: 3–24.
- Merizow, J. (ed.), and Associates. 1990. *Fostering critical reflection in adulthood: A guide to transformative and emancipatory learning*. San Francisco, CA: Jossey-Bass.
- Mills, R. 2008. "It's just a nuisance": Improving college student reflective journal writing. *College Student Journal* 42, no. 2: 684–90.
- Minott, M.A. 2008. Valli's typology of reflection and the analysis of pre-service teachers' reflective journals. *Australian Journal of Teacher Education* 33, no. 5: 55–65.
- Moon, J.A. 2006. *Learning journals: A handbook for reflective practice and professional development*, 2nd ed. London: Routledge.
- O'Connell, T.S., and J.E. Dymont. 2003. Effects of a journaling workshop on participants in university outdoor education field courses: An exploratory study. *Journal of Experiential Education* 26, no. 2: 75–87.
- O'Connell, T.S., and J.E. Dymont. 2004. Journals of post secondary outdoor recreation students: The result of a content analysis. *Journal of Adventure Education and Outdoor Learning* 4, no. 2: 159–72.
- O'Connell, T.S., and J.E. Dymont. 2006. Reflections on using journals in higher education: A focus group discussion with faculty. *Assessment and Evaluation in Higher Education* 31, no. 6: 671–91.
- Paget, T. 2001. Reflective practice and clinical outcomes: Practitioners' views on how reflective practice has influenced their clinical practice. *Journal of Clinical Nursing* 10, no. 2: 204–14.
- Plack, M.M., M. Driscoll, S. Blissett, R. McKenna, and T.P. Plack. 2005. A method for assessing reflective journal writing. *Journal of Allied Health* 34, no. 4: 199–208.
- Powell, J.H. 1989. The reflective practitioner in nursing. *Journal of Advanced Nursing* 14: 824–32.
- Richardson, G., and H. Maltby. 1995. Reflection-on-practice: Enhancing student learning. *Journal of Advanced Nursing* 22, no. 2: 235–42.
- Scanlon, J.M., and W.M. Chernomas. 1997. Developing the reflective teacher. *Journal of Advanced Nursing* 25, no. 6: 1138–43.
- Schön, D.A. 1983. *The reflective practitioner*. New York: Basic Books.
- Schön, D.A. 1987. *Educating the reflective practitioner*. San Francisco, CA: Jossey-Bass.
- Shor, I. 1992. *Empowering education: Critical teaching for social change*. Chicago, IL: University of Chicago Press.
- Spalding, E., and A. Wilson. 2002. Demystifying reflection: A study of pedagogical strategies that encourage reflective journal writing. *Teachers College Record* 104, no. 7: 1393–421.
- Sumsion, J., and A. Fleet. 1996. Reflection: Can we assess it? Should we assess it? *Assessment and Evaluation in Higher Education* 21, no. 2: 121–31.

- Thorpe, K. 2004. Reflective learning journals: From concept to practice. *Reflective Practice* 5, no. 3: 327–43.
- Valli, L. 1997. Listening to other voices: A description of teacher reflection in the United States. *Peabody Journal of Education* 72, no. 1: 67–88.
- Wessel, J., and H. Larin. 2006. Change in reflection of physiotherapy students over time in clinical placements. *Learning in Health and Social Care* 5, no. 3: 119–32.
- Williams, R.M., and G. Sundelin. 2000. Assessing the reliability of reflective journal writing. *Journal of Physical Therapy Education* 14, no. 2: 23–6.
- Williams, R.M., J. Wessel, M. Gemus, and E. Foster-Seargeant. 2002. Journal writing to promote reflection by physical therapy students during clinical placements. *Physiotherapy Theory and Practice* 18, no. 1: 5–15.
- Wong, F.K.Y., D. Kember, L.Y.F. Chung, and L. Yan. 1995. Assessing levels of reflection from reflective journals. *Journal of Advanced Nursing* 22, no. 1: 48–57.
- Wong, F.K.Y., A.Y. Loke, M. Wong, H. Tse, E. Kan, and D. Kember. 1997. An action research study into the development of nurses as reflective practitioners. *Journal of Nursing Education* 34, no. 10: 476–81.